

Structural Shocks and Jump Dynamics in the Crude Oil Market

Abstract

This paper studies how supply demand macroeconomic and speculative shocks shape jumps and tail risk in the WTI crude oil futures market. Using high frequency intraday data we decompose realized variance into continuous volatility and discontinuous jumps to separate extreme movements from regular price dynamics. We apply variance decomposition jump regressions and quantile regressions to evaluate the role of structural shocks. Results show that supply and speculative shocks amplify volatility and jump intensity while macroeconomic and consumption shocks reduce both. Quantile regressions reveal that macroeconomic shocks lower downside extremes and compress the upper tail and that speculative shocks raise left tail risk. Including jump dynamics in a quantile based Value at Risk framework improves predictive accuracy over Historical Simulation GARCH and non jump CAViaR models. The findings highlight the role of structural shocks in tail risk and the value of linking fundamentals to risk management.

1 Introduction

West Texas Intermediate (WTI) crude oil is a global benchmark for energy markets and a central reference for pricing in both physical and derivative oil transactions. As both a critical production input and a widely traded financial asset, WTI price movements influence and are influenced by economic activity, inflation dynamics, and investment decisions worldwide (?).

This dual role creates an endogenous feedback loop between the oil market and the broader economy.s

The determinants of WTI price dynamics extend well beyond market microstructure factors such as inventories and storage costs. Broader macroeconomic forces and structural shocks in the global oil market also play a critical role. Empirical evidence shows that oil prices respond to distinct types of demand and supply disturbances, including flow supply shocks, aggregate demand shocks linked to the global business cycle, and oil market-specific or speculative demand shocks that reflect changes in expectations about future supply-demand balances (???). These shocks propagate through complex market mechanisms, generating both gradual, continuous price adjustments and sudden, discontinuous jumps that can disproportionately affect risk exposure and market stability.

While the structural decomposition of oil price movements provides valuable economic insights, a parallel strand of research has emphasized the statistical modeling of crude oil prices and returns, with particular focus on the role of jumps. The existence of jumps in asset markets is well established: ? provide early evidence of discontinuities in both returns and volatility, while ? show that jumps are statistically significant, accounting for roughly 7% of total asset price variance.

To capture these features, commodity price dynamics are often modeled using stochastic process frameworks that explicitly incorporate jumps, including stochastic volatility, jump-diffusion, and regime-switching models (???). Within these frameworks, jumps in the return process are recognized as critical for derivative pricing, risk management, and market microstructure analysis (???).

The economic origins of jumps in crude oil prices remain less studied compared with the extensive literature on oil price volatility. Prior research has examined the relationship between volatility and speculative activity (??) as well as the influence of macroeconomic conditions (??) and other market variables such as equity volatility or inventory constraints (??). In contrast the drivers of price jumps are less well understood. ? show that crude oil tail risks are linked to demand and supply fundamentals with demand shocks driving downside risks in the short run and supply shocks shaping upside risks over longer horizons. This evidence indicates that jumps are not random anomalies but are systematically related

to macroeconomic conditions financial uncertainty and oil market fundamentals.

This paper addresses this gap by bridging the structural decomposition of oil market shocks with statistical jump models, providing a more nuanced understanding of the transmission of economic shocks into extreme market movements. We focus on the jump components of crude oil price dynamics and examine how they are systematically related to underlying fundamental shocks. We further extend the analysis to risk management by assessing the extent to which incorporating jump components enhances the performance of market risk measurement.

The empirical analysis yields four main findings. First extreme oil price risks are primarily driven by jumps which cluster during crises and exhibit pronounced directional asymmetries. Second structural shocks shape both continuous volatility and jump variation in distinct ways. Supply shocks amplify overall variation with especially strong effects on positive jumps. Macroeconomic and consumption shocks dampen both continuous and jump components. Speculative shocks reduce average jump sizes more strongly on the downside while shifting intensity toward more frequent positive jumps. Third the impact of structural shocks on returns is heterogeneous across the distribution. Macroeconomic shocks mitigate downside risk but compress upside extremes. Supply shocks depress central returns yet fade in the tails. Consumption shocks raise lower tail returns and speculative shocks matter mainly at the center. Finally, incorporating jump variation into Value-at-Risk models improves predictive performance where persistence and clustering effects make lagged jump information informative.

The remainder of the paper is organized as follows. Section ?? reviews the related literature. Section ?? presents the empirical methodology. Section ?? describes the data. Section ?? discusses the empirical findings. Section ?? concludes.

2 Literature Review

One strand of research investigates the determinants of crude oil price dynamics, linking price movements to both macroeconomic conditions and market-specific fundamentals. The structural VAR tradition pioneered by ? decomposes oil price changes into flow supply

shocks, aggregate demand shocks tied to the global business cycle, and oil-specific demand shocks reflecting expectations. Subsequent work has emphasized the role of inventories and speculative trading as amplification mechanisms (?) and has introduced Bayesian approaches that relax strict identifying assumptions while preserving economically meaningful interpretations (?). This strand has concentrated on explaining aggregate price dynamics, with less attention to how structural shocks affect discontinuities or extreme market outcomes.

Beyond direct price dynamics, another strand of research focuses on the time-varying volatility of crude oil returns. Volatility in this context may correspond to integrated volatility when returns follow a pure diffusion process, or to quadratic variation when returns also contain discontinuous jumps. This literature investigates how macroeconomic conditions (e.g., ????) and speculative activity (e.g., ??) shape crude oil volatility. However, studies rely on GARCH-type or realized volatility measures that aggregate both the continuous (diffusive) and discontinuous (jump) components of the return process, leaving the determinants of the isolated jump component relatively underexplored.

In continuous-time models of asset prices, there is growing evidence that jumps in returns and volatility are closely intertwined, suggesting common underlying drivers. In equity markets, ? show that price and volatility jumps often co-occur and are negatively related, with downward price jumps typically accompanied by upward volatility jumps. Consistent with this, ? develop a nonparametric test for price–volatility co-jumps and document nonlinear dependence. Extending these insights to commodities, ? use copula methods to show that negative crude oil price jumps are strongly associated with positive volatility jumps, while ? demonstrate within a GARCH-jump framework that accounting for time-varying jump intensity significantly improves forecasts of WTI and Brent volatility.

Turning to the drivers of jump components, early evidence from ? indicates that jumps frequently coincide with macroeconomic announcements, suggesting they are information-driven rather than random noise. Building on this view, ? show in currency markets that both jump sizes and intensities are systematically related to macroeconomic conditions, providing a foundation for studying the economic origins of discontinuities. In the case of crude oil, ? find that jump intensity increases sharply during periods of financial turmoil, with oil volatility, the VIX, and economic policy uncertainty (EPU) explaining much of its time variation. More

recently, ? link these discontinuities to structural oil market fundamentals, showing that demand shocks primarily drive downside tail risks in the short run, whereas supply shocks dominate upside risks over longer horizons. Overall, the evidence demonstrates that jumps are not random anomalies but are systematically tied to macroeconomic conditions, financial uncertainty, and market-specific fundamentals.

Given that jumps are not purely random but are at least partly driven by fundamental supply and demand factors, this study links jump variation, as modeled within a continuous-time stochastic process, to the underlying economic determinants of oil prices, thereby providing a more comprehensive understanding of the dynamics of discontinuities.

Risk assessment in crude oil markets has been widely examined through Value-at-Risk (VaR), which is defined as the $\alpha\%$ loss threshold over a holding period h (e.g., ?). Yet few studies directly model VaR with jumps. Approaches based on volatility models and distributional assumptions sometimes incorporate jumps as exogenous variables, as in GARCH-MIDAS frameworks (e.g., ?). Other approaches based on the extreme value theorem capture extreme price movements indirectly by modelling the distribution of the tails.

Motivated by these limitations, the present study employs quantile-based methods, specifically the asymmetric CAViaR model of ?, to examine the role of jump components in VaR estimation. This approach allows explicit evaluation of both long and short positions, or equivalently the left and right tails, thereby addressing asymmetric tail risk in crude oil markets (see also ?).

3 Methods

3.1 Jump Detection

Two broad strands of the literature address the problem of testing for the existence of jumps in continuous-time models (e.g., ?). The first strand focuses on the transition density or other structural properties of continuous-time models. These approaches examine the probabilistic law governing the process itself, thereby testing for the presence of jumps in the model class rather than in a single realized sample path (e.g., ????). Such tests are particularly relevant

for model selection, for instance, when deciding between pure diffusion and jump-diffusion models in option pricing.

The second strand relies on observed high-frequency sample paths to decompose total variation into continuous and jump components, typically using scaling laws, truncation methods, or comparisons of realized measures (e.g., ?????). These pathwise tests are primarily designed for empirical jump detection and measurement.

Similar to ?, this study adopts the Barndorff-Nielsen framework by decomposing daily return variance into continuous and jump components, thereby identifying jump events without imposing parametric assumptions on the data-generating process.

Assuming the jump-diffusion process follows:

$$dp_t = \mu_t dt + \sigma_t dW_t + dJ_t \quad (1)$$

where p_t is the logarithmic price (e.g., WTI crude oil price), μ_s is the drift process, σ_s is the continuous component of variance, and W_s is the Brownian motion component of variance, J_t is a pure jump process.

The corresponding quadratic variation (QV) is

$$[p]_t = \int_0^t \sigma_s^2 ds + \sum_{0 < s < t} (\Delta p_s)^2, \quad (2)$$

where $\Delta p_s = p_s - p_{s-}$ if a jump happens.

Given Intraday returns at a fixed interval Δ (e.g., 5-minute sampling) are computed as

$$r_{t_i} = p_{t_i} - p_{t_{i-1}}, \quad (3)$$

The quadratic variation (QV) of asset prices can be consistently estimated using the realized variance (RV), defined as the sum of squared intraday returns within a day. The continuous component of QV can in turn be approximated by the bipower variation (BV). Formally (see ??):

$$RV_t = \sum_{i=1}^n r_{t,i}^2 \xrightarrow{p} [p]_t, \quad \text{as } n \rightarrow \infty, \quad (4)$$

$$BV_t = \frac{n}{n-1} \cdot \frac{\pi}{2} \sum_{i=2}^n |r_{t,i}| \cdot |r_{t,i-1}| \xrightarrow{p} \int_0^t \sigma_s^2 ds, \quad (5)$$

where $r_{t,i}$ denotes the i -th intraday return on day t , and n is the total number of intraday observations. The difference $RV_t - BV_t$ then provides a nonparametric estimate of the jump component of price variation (JV).

$$JV = RV - BV \xrightarrow{p} \sum_{0 < s < t} (\Delta p_s)^2 \quad (6)$$

Following the upside and downside realised semivariance measures proposed by ?, the realised variance can be decomposed into the sum of squared positive and negative intraday returns.

$$RS^+ = \sum_{i=1}^n r_i^2 \mathbb{I}(r_i > 0) \xrightarrow{p} \frac{1}{2} \int_0^t \sigma_s^2 ds + \sum_{0 \leq s \leq t} (\Delta p_s)^2 \mathbb{I}(\Delta p_s > 0) \quad (7)$$

$$RS^- = \sum_{i=1}^n r_i^2 \mathbb{I}(r_i < 0) \xrightarrow{p} \frac{1}{2} \int_0^t \sigma_s^2 ds + \sum_{0 \leq s \leq t} (\Delta p_s)^2 \mathbb{I}(\Delta p_s < 0) \quad (8)$$

It follows from the decomposition that the positive and negative jump variation can be written as:

$$JV^+ = \sum_{0 \leq s \leq t} (\Delta p_s)^2 \mathbb{I}(\Delta p_s > 0) \xrightarrow{p} RS^+ - \frac{1}{2} \int_0^t \sigma_s^2 ds, \quad (9)$$

$$JV^- = \sum_{0 \leq s \leq t} (\Delta p_s)^2 \mathbb{I}(\Delta p_s < 0) \xrightarrow{p} RS^- - \frac{1}{2} \int_0^t \sigma_s^2 ds. \quad (10)$$

The use of signed indicator functions allows the extraction of directional jump components, enabling separate modelling of left- and right-tail jump variations.

To account for time-varying volatility and small-sample properties, the tripower quarticity (TQ) is employed. It is defined as

$$TQ_t = n \cdot \frac{n}{n-2} \cdot \mu_{4/3}^{-3} \sum_{i=3}^n |r_{t,i}|^{4/3} \cdot |r_{t,i-1}|^{4/3} \cdot |r_{t,i-2}|^{4/3}, \quad (11)$$

where $\mu_{4/3} = 2^{2/3} \cdot \frac{\Gamma(7/6)}{\Gamma(1/2)}$. This statistic provides a robust estimate of the variation in

volatility and improves the stability of the jump test under heteroskedasticity.

The standardized test statistic for jump detection is given by

$$ZJ_t = \frac{(RV_t - BV_t)/RV_t}{\sqrt{\left(\frac{\pi}{2}\right)^2 + \pi - 5 \cdot \max\left(1, \frac{TQ_t}{BV_t^2}\right)}/n}. \quad (12)$$

A significant jump is identified when $|ZJ_t|$ exceeds the critical value from the standard normal distribution corresponding to the chosen significance level α .

The magnitude of the jump component is estimated as

$$\hat{J}_t = \text{sign}\left(\sum_{i=1}^n r_{t_i}\right) \times \sqrt{\max(RV_t - BV_t, 0)} \times \mathbb{I}\left(|ZJ_t| > z_{1-\alpha/2}\right). \quad (13)$$

This method captures both the magnitude and direction of jumps, offering a more detailed view of tail risk than volatility measures alone can provide.

For empirical implementation, daily jump statistics are aggregated at the monthly level to construct a panel of risk indicators. Let $\mathcal{D}_m$ denote the set of trading days in month m . The following monthly aggregates are computed:

$$\text{JumpCount}_m = \sum_{t \in \mathcal{D}_m} \mathbb{I}\left(|ZJ_t| > z_{1-\alpha/2}\right), \quad (14)$$

where $\mathbb{I}(\cdot)$ equals one when a jump is detected on day t . This measure captures the frequency of jump occurrences within each month.

To assess directional jump risk, we decompose the count and the size of jumps by sign. First, directional counts:

$$\text{JumpCount}_m^+ = \sum_{t \in \mathcal{D}_m} \mathbb{I}\left(|ZJ_t| > z_{1-\alpha/2}, \hat{J}_t > 0\right) \quad (15)$$

$$\text{JumpCount}_m^- = \sum_{t \in \mathcal{D}_m} \mathbb{I}\left(|ZJ_t| > z_{1-\alpha/2}, \hat{J}_t < 0\right) \quad (16)$$

Second, directional sizes. Let $J_t^+ \equiv \max(\hat{J}_t, 0)$ and $J_t^- \equiv \max(-\hat{J}_t, 0)$ denote the positive and

negative parts of the signed jump size estimator $\hat{J}_t$. We aggregate daily sizes to the month:

$$\text{JumpSize}_m^+ = \sum_{t \in \mathcal{D}_m} J_t^+ \mathbb{I}(|ZJ_t| > z_{1-\alpha/2}) \quad (17)$$

$$\text{JumpSize}_m^- = \sum_{t \in \mathcal{D}_m} J_t^- \mathbb{I}(|ZJ_t| > z_{1-\alpha/2}) \quad (18)$$

These aggregates capture the monthly contributions of positive and negative jump magnitudes, complementing the directional counts. Taken together, the monthly jump size and count measures provide a basis for analyzing directional jump risk and attributing observed jump activity to underlying structural shocks.

3.2 Jump Tail Distribution

The probability of observing a large loss can be decomposed into two mutually exclusive cases: the absence or occurrence of jumps,

$$\Pr(r_t < -x) = \Pr(r_t^{(c)} < -x, N_t = 0) + \Pr(r_t^{(c)} + J_t < -x, N_t > 0). \quad (19)$$

When no jump occurs, the tail probability of the continuous component satisfies

$$\Pr(r_t^{(c)} < -x) = O\left(e^{-x^2/(2\sigma_t^2)}\right), \quad x \rightarrow \infty, \quad (20)$$

which decays exponentially under Gaussian diffusive dynamics.

When a jump occurs, the total return combines diffusion and a (possibly large) discontinuous jump. Under Assumptions A1–A2 of ?, the large-jump compensator follows a regularly varying tail with index $\alpha_- > 0$, whereas the continuous part remains light-tailed:

$$\Pr(r_t^{(c)} + J_t < -x) = C x^{-\alpha_-} + o(x^{-\alpha_-}), \quad x \rightarrow \infty, \quad (21)$$

where the remainder $o(x^{-\alpha_-})$ subsumes all components that decay faster than the Pareto term, including $\Pr(r_t^{(c)} < -x) = O\left(e^{-x^2/(2\sigma_t^2)}\right) = o(x^{-\alpha_-})$. Consequently,

$$\Pr(r_t < -x) \sim C x^{-\alpha_-}, \quad x \rightarrow \infty, \quad (22)$$

Hence, the polynomially decaying jump tail is asymptotically thicker than the exponentially decaying Gaussian tail, and thus dominates the overall return distribution in the far left tail. This theoretical result aligns with the empirical evidence in ?, who emphasize that extreme market losses are primarily driven by rare, discontinuous jumps rather than continuous volatility fluctuations.

Building on this framework, ? apply Extreme Value Theory (EVT) and estimate the tail decay parameters nonparametrically using the Peaks-Over-Threshold (POT) method. Let J_t denote the jump size at time t . For a sufficiently high threshold u , the conditional excess distribution is

$$F_u(y) = \Pr(J_t - u \leq y \mid J_t > u), \quad y > 0. \quad (23)$$

By the Pickands–Balkema–de Haan theorem, $F_u(y)$ converges to a Generalized Pareto Distribution (GPD),

$$F_u(y) \approx G_{\xi, \beta}(y) = 1 - \left(1 + \frac{\xi y}{\beta}\right)^{-1/\xi}, \quad y > 0, \quad (24)$$

where ξ is the shape parameter (tail index) and β is the scale parameter. The estimated $\hat{\xi}$ directly measures tail thickness: $\hat{\xi} > 0$ implies heavy (regularly varying) tails, $\hat{\xi} = 0$ corresponds to an exponential tail, and $\hat{\xi} < 0$ implies bounded jump sizes.

3.3 Quantile-Based VaR

To capture the dynamics of tail risk beyond volatility-based approaches, this study adopts the Conditional Autoregressive Value-at-Risk (CAViaR) framework of ?. In particular, we employ the Asymmetric Slope specification, which directly models the evolution of return quantiles without imposing parametric distributional assumptions. This specification allows for differential responses to positive and negative return shocks and can be extended to incorporate exogenous variables.

Formally, for a given quantile level $\tau \in (0, 1)$, the τ -th conditional quantile of the return

r_{t+1} is expressed as

$$\text{VaR}_{t+1}(\tau) = \beta_0 + \beta_1 \text{VaR}_t(\tau) + \beta_2 \max(0, -r_t) + \beta_3 \max(0, r_t). \quad (25)$$

We extend this model by including jump variations as exogenous regressors:

$$\text{VaR}_{t+1}(\tau) = \beta_0 + \beta_1 \text{VaR}_t(\tau) + \beta_2 \max(0, -r_t) + \beta_3 \max(0, r_t) + \gamma_1 \text{JV}_t^+ + \gamma_2 \text{JV}_t^-, \quad (26)$$

where JV_t^+ and JV_t^- denote the positive and negative jump variations in month t , respectively. These measures are constructed from high-frequency data as outlined in the preceding subsection and capture the influence of discontinuous price movements on tail risk.

The parameters $\theta = (\beta_0, \beta_1, \beta_2, \beta_3, \gamma_1, \gamma_2)$ are estimated by minimizing the quantile regression loss function

$$\hat{\theta} = \arg \min_{\theta} \sum_{t=1}^T \rho_{\tau}(r_{t+1} - \text{VaR}_{t+1}(\tau; \theta)), \quad (27)$$

where $\rho_{\tau}(u) = u \cdot (\tau - \mathbb{I}\{u < 0\})$ is the standard check function.

In this specification, the autoregressive term captures persistence in conditional quantiles, the asymmetric slope terms reflect nonlinear responses to positive and negative return shocks, and the jump variation terms introduce information about discontinuous movements. By incorporating JV_t^+ and JV_t^- , the model explicitly connects tail risk forecasts to realized jump dynamics, thereby allowing an assessment of the role of jumps in shaping extreme downside and upside risks in crude oil markets.

4 Data

The 5-minute intraday continuous nearest-to-maturity trading data of WTI crude oil price are collected from Chicago Mercantile Exchange (CME). The data covers the period from 1st of January 2008 to 1st of July 2025. The monthly oil demand and supply shocks are taken from ?, who employ a Bayesian framework to identify and interpret structural shocks in the global oil market. For the purpose of this study, we directly use the shocks data provided on

the author’s personal website, rather than reconstructing them from scratch.

[Insert Table ?? about here.]

Table ?? reports the summary statistics of daily crude oil returns over the full sample and selected volatile subperiods. The selected subperiods are: Global Financial Crisis (2008-01 to 2009-06); 2014 Oil Price Collapse (2014-06 to 2016-02); 2018 Trade War (2018-03 to 2019-12); COVID-19 Pandemic (2020-02 to 2021-12); and 2022 Ukraine Crisis (2022-02 to 2023-02)¹.

Across the entire sample, returns display a near-zero mean, consistent with the random-walk property of asset prices at high frequency. Annualized volatility rises sharply during turbulent episodes, reflecting the oil market’s sensitivity to geopolitical and macroeconomic shocks. The distribution of returns is generally leptokurtic, with significant excess kurtosis pointing to fat tails and frequent extreme price changes. Skewness is positive in most cases, indicating that large upward movements tend to dominate in normal times and many volatility episodes. However, during the U.S.-China trade war and the Russia-Ukraine war, the skewness turns negative, suggesting that downside risks and large price drops dominated during these crises. This asymmetry in the return distribution across different regimes highlights the importance of accounting for both fat tails and sign-dependent tail risks when modeling crude oil price dynamics.

[Insert Figure ?? about here.]

Figure ?? presents the monthly demand- and supply-driven shocks (Panel (a)) together with the corresponding monthly returns (Panel (b)). Supply, macro, and speculation shocks fluctuate moderately around zero, whereas consumption shocks display much larger swings, particularly during crisis periods such as 2008–2009, 2020, and 2021–2022. Panel (b) further shows pronounced volatility clustering in returns. The coincidence of large structural shocks with extreme return episodes underscores the pivotal role of these shocks in driving oil price dynamics.

[Insert Figure ?? about here.]

¹Period definitions follow IMF and World Bank reports on major oil and financial crises. Results are robust to small changes in start or end dates.

Figure ?? illustrates the distributional properties of daily WTI crude oil returns. Panel (a) shows the return distribution using both a histogram and kernel density estimation (KDE), compared against a fitted normal distribution. The return series is more peaked around the mean and exhibits heavier tails relative to the normal benchmark, highlighting the leptokurtic nature of crude oil returns. Panel (b) presents the corresponding QQ plot, where notable deviations from the 45-degree line are observed in the tails. This further confirms that the return distribution departs from normality, with extreme returns occurring more frequently than predicted by a Gaussian distribution. Together, these plots underscore the importance of accounting for fat tails and non-normal features when modeling oil price dynamics.

5 Results

5.1 Jump Activity

Table ?? reports the frequency and direction of oil price jumps during major market turmoil episodes. The share of jump days differs considerably across crises. The 2014 Oil Price Collapse stands out, with more than one third of trading days identified as jump days, followed by the 2018 Trade War at just over 31%. By contrast, the COVID-19 pandemic, the 2022 Ukraine crisis, and the Global Financial Crisis exhibit lower jump intensities, with roughly 23%, 23%, and 25% of days classified as jump days, respectively.

[Insert Table ?? about here.]

This heterogeneity reflects the nature of the underlying disruptions: the 2014 collapse and trade tensions involved protracted uncertainty and continuous reassessment of supply-demand conditions, generating frequent discontinuous adjustments (?), whereas the Global Financial Crisis, the pandemic, and the Ukraine war produced sharp one-off shocks and policy responses that limited the number of distinct jump events despite their severity.

The decomposition into positive and negative jumps highlights notable asymmetries. These are most pronounced during the 2014 collapse, the COVID-19 pandemic, and the Ukraine crisis. In both the 2014 collapse and the Ukraine crisis, negative jumps dominate, reflecting the adverse supply shocks and geopolitical disruptions that placed persistent

downward pressure on prices and amplified downside risk. By contrast, the COVID-19 pandemic displays a higher frequency of positive jumps. Following the sharp collapse in early 2020, prices rebounded as unprecedented policy interventions, coordinated production cuts, and progress on vaccine development restored market confidence, resulting in more frequent upward corrections.

The results demonstrate that jump activity is both crisis-specific and directionally asymmetric. Episodes associated with supply-side disruptions or geopolitical shocks tend to generate a higher prevalence of negative jumps, whereas financially driven crises exhibit a more balanced distribution of jump directions. This evidence underscores the importance of distinguishing between positive and negative jumps when assessing the impact of structural shocks on oil market dynamics.

[Insert Figure ?? about here.]

Figure ?? illustrates the time-series evolution of oil price jump sizes and counts, both in aggregate and decomposed into positive and negative components. Panel (a) shows that total jump sizes reach local peaks during major episodes of market stress, including the 2008-09 Global Financial Crisis, the 2014-16 shale oil boom and subsequent price collapse, the COVID-19 pandemic, and the 2022 Ukraine crisis. Panel (b) highlights directional asymmetries: positive jump sizes generally exceed negative jump sizes over the full sample, with the dominance of positive jumps particularly evident during the Global Financial Crisis and the COVID-19 episode. This could be due to large-scale policy interventions, liquidity injections, and coordinated supply adjustments, which fostered sharp upward corrections in oil prices and produced outsized positive jumps following the initial downturns (e.g., see ?). By contrast, the 2014-16 collapse exhibits a more clustered jump pattern, consistent with continuous reassessment of supply-demand conditions and repeated discontinuous price adjustments. Panels (c) and (d) present total jump counts and their decomposition by sign. In contrast to jump sizes, jump counts display more persistent clustering, with positive and negative counts showing strong temporal dependence in Panel (d). This behavior resembles volatility clustering and provides empirical support for modeling jump arrivals with self-exciting processes such as Hawkes in the context of commodity price dynamics.

By documenting that both the magnitude and frequency of jumps vary across crises and display directional asymmetries, the analysis underscores that jump activity is uneven over time, responding to structural shocks and shaped by whether disruptions stem from supply or demand conditions.

To understand the role of jumps in shaping tail risk, we analyze the extreme return distribution using the Peaks-over-Threshold (POT) framework. Following ?, the jump compensator for large jumps behaves like a probability measure, regardless of temporal variation in jump intensity. This property enables a nonparametric characterization of jump tails via decay parameters estimated under POT. Table ?? reports Generalized Pareto Distribution (GPD) fits for both returns and jump sizes. The estimated shape parameters, ξ , show that jump-size tails are substantially heavier—with $\xi \approx 0.58$ – 0.68 (tail indices $\alpha = 1/\xi \approx 1.5$ – 1.7)—than return tails, which exhibit $\xi \approx 0.37$ – 0.59 (i.e., $\alpha \approx 1.7$ – 2.7). These results confirm that the heaviness of the extreme return distribution is largely inherited from jumps, consistent with ?, who document heavy-tailed jump distributions that dominate stock-return tails.

[Insert Table ?? about here.]

It has been established that the tail risk of crude oil returns is primarily driven by jumps. Building on this insight, we next examine how structural supply and demand shocks shape the dynamics of jump activity.

5.2 Continuous and Jump Components of Variance

Table ?? presents regression estimates of realized variance and its decomposition into jump and continuous components on structural shocks. The variation measures display characteristics consistent with an $I(1)$ process, as regressions on their one-period lag yield coefficients close to unity. To achieve stationarity and mitigate the risk of spurious regression, we apply first differencing. The resulting specification is thus cast within an $AR(1)$ framework augmented with exogenous covariates, corresponding to an ARX model.

[Insert Table ?? about here.]

In terms of supply-side shocks, the coefficients are positive and statistically significant across all measures of realized variation. The coefficient on $\Delta \log RV_t$ is 0.1008, while the effect on jump variation is of similar magnitude at 0.0952. The impact is particularly pronounced for positive jump semivariance (JV_t^+), with an estimate of 0.1205, indicating that supply shocks raise the positive jump variations. By contrast, the coefficient on negative jump semivariance is smaller, though still positive, at 0.0944, suggesting that supply shocks also contribute to downside jumps, albeit to a lesser extent.

The results are somewhat counterintuitive. Positive supply shocks typically exert downward pressure on prices, which would suggest larger variation in downside jumps. However, the empirical evidence indicates that under large positive shocks, the variation in positive jumps not only exceeds that of negative jumps but is also statistically more significant. One possible explanation is that following a positive supply shock, the market may overreact (e.g., see ??), thereby amplifying the short-term rebound in prices after the shock. The results in the first row of Table ?? imply that supply disturbances amplify both return and jump variability, with asymmetrically stronger effects on the upper tail, as reflected in the comparison between $\Delta \log RS_t^+$ and $\Delta \log RS_t^-$ as well as between $\Delta \log JV_t$ and $\Delta \log JV_t^+$.

Macroeconomic shocks, capturing global demand for all industrial commodities, display consistently negative and highly significant coefficients across all specifications. Comparing BV and JV in columns 3 and 4, which capture continuous diffusive variation and jump variation, respectively, reveals that demand shocks exert a stronger impact on jump variation (-0.1636) than on diffusive variation (-0.1154). Moreover, the coefficients for JV^+ and JV^- are similar, indicating largely symmetric effects. The results suggest that favorable macroeconomic conditions reduce both total variance and jump-related risk, even if they contribute to upward price pressures, whereas adverse macro shocks tend to push prices downward while simultaneously amplifying both diffusive volatility and jump tail risk.

Consumption shocks, capturing shifts in demand for crude oil itself, also exert negative and statistically significant effects across all measures of variation. The coefficients are consistently around 0.06, indicating a uniform influence on both continuous volatility and jump components. This pattern suggests that stronger consumption demand stabilizes return dynamics broadly, reducing overall variability. Speculative shocks, reflecting shifts

in expectations about future oil demand, also enter with a negative sign. A comparison of RS^+ (-0.0912) versus RS^- (-0.0700) and JV^+ (-0.1400) versus JV^- (-0.0374) indicates that positive jump components are more strongly influenced by speculative shocks than their negative counterparts.

Comparing the coefficients across supply and demand shocks reveals a clear asymmetry. Positive supply shocks are associated with higher variance and greater jump risk, whereas demand shocks exert the opposite effect. Specifically, a positive supply shock and a negative demand shock both depress prices and are accompanied by increases in continuous and jump variation. Conversely, a negative supply shock and a positive demand shock both raise prices and are associated with declines in continuous and jump variation. This asymmetry can be rationalized by the cyclical nature of volatility: oil price increases often coincide with periods of economic expansion, when market participants are more optimistic and volatility subsides (???). By contrast, oil price declines are frequently interpreted as signals of economic contraction, a context in which uncertainty intensifies and volatility rises as market participants become more pessimistic and risk-averse.

5.3 Jump Sizes and Intensities

In the previous subsection, we showed that supply and demand shocks influence jump variation without distinguishing the underlying channels. This subsection examines more directly whether these effects operate through changes in jump size or in jump intensity. Jump sizes, whether positive or negative, are taken in absolute value, and their logarithms are regressed on supply and demand shocks. For the jump count regressions, we address the limitation of the standard Poisson model, which imposes equality of the conditional mean and variance. To account for overdispersion that arises from the heterogeneous timing of jumps, since some days are inherently more prone to multiple jumps, we employ the Negative Binomial specification, which relaxes this restriction by allowing the variance to exceed the mean.

[Insert Table ?? about here.]

Table ?? presents the estimated effects of structural shocks on jump sizes and jump counts. The first two columns report log-linear regressions of jump sizes on structural shocks. The

results show that both supply and macroeconomic shocks exert insignificant influence on jump sizes. Consumption shocks affect only negative jump sizes, with higher consumption shocks associated with smaller negative jumps. In contrast, speculative shocks display a consistently negative and significant effect on jump sizes. This finding suggests that increases in speculative oil demand tend to reduce the magnitude of jumps. Moreover, the effect is stronger for negative jump sizes (-0.2946) than for positive jump sizes (-0.2197), indicating that heightened speculative demand, by driving prices upward, disproportionately mitigate the severity of downward jumps. This finding supports the literature(??) that speculation play little role in destabilize the market from the aspect of jumps.

The last two columns report Negative Binomial regressions of jump counts on structural shocks. Relative to the size regressions, most coefficients are statistically significant at the one percent level (with the exceptions of the macro shock for negative jumps and the consumption shock for positive jumps), indicating that demand and supply information primarily operates through the intensity of jump arrivals rather than their magnitude. This finding aligns with ?, who show that during market turmoil jump frequency changed sharply while average sizes remained stable.

Supply shocks exhibit clear asymmetry as the coefficient on positive jump counts is negative (-0.3975) whereas the coefficient on negative jump counts is positive (0.2462). This pattern is consistent with the price impact of favorable supply news where unexpected increases in supply tend to depress prices and therefore raise the incidence of left tail negative jumps while curbing right tail positive jumps.

Macroeconomic shocks primarily influence the right tail, as expansions are associated with a significant decline in positive jump intensity (-0.5446), while their effect on negative jump intensity is small and statistically insignificant. This pattern is consistent with the evidence that stronger macro fundamentals coincide with lower overall jump and continuous variation. Favorable macroeconomic conditions can stabilize the market by reducing the frequency of large discontinuous moves rather than altering their average jump size.

Unlike macro shocks, consumption shocks load primarily on the left tail. An unexpected strengthening in consumption demand is associated with a higher intensity of negative jumps, as evidenced by a positive and significant coefficient for the negative jump count, while the

effect on positive jump counts remains weak. One interpretation is that oil-specific demand shocks, often linked to precautionary motives, raise prices but also heighten uncertainty. Market participants may overreact to such signals, increasing the likelihood of downside corrections and thereby amplifying the incidence of negative jumps.

It is worth highlighting the role of speculative shocks, which are significant across all specifications in the last row of Table ???. Increases in demand driven by market expectations reduce jump sizes in both directions, while simultaneously raising the intensity of positive jumps and lowering the intensity of negative jumps. This asymmetric pattern is particularly pronounced on the left tail, where the decline in negative jump intensity (-1.3571) is much larger than the corresponding increase in positive jump intensity (0.7079). These results provide nuanced evidence that speculative behavior tends to mitigate rather than amplify discontinuous price movements.

5.4 Quantile Analysis of Returns

To complement the preceding analysis of jump intensities, we estimate quantile regressions of returns on structural shocks. This exercise highlights how shocks covary with returns across the conditional distribution, thereby uncovering heterogeneity in transmission channels with particular attention to the tails. Table ?? reports coefficient estimates and robust t -statistics for quantiles ranging from the 1st to the 99th percentile.

[Insert Table ?? about here.]

At the center of the distribution (q_{50}), all structural shocks significantly affect returns. Supply shocks enter with a negative coefficient (-0.1121), consistent with downward pressure on prices, while demand-side shocks are positive: speculative (0.0954), consumption (0.0773), and macroeconomic (0.0740). These findings are consistent with the literature on structural supply and demand shocks in the oil market (???).

On the left tail (q_{01} and q_{05}), demand-side shocks dominate. Macroeconomic shocks are particularly strong, with coefficients of 1.0087 and 0.5155, both highly significant, followed by consumption shocks (0.2526 and 0.1710). Speculative shocks are positive but weaker, significant only at the 5th percentile (0.1335). These results indicate that, in extreme

downturns, stronger macroeconomic and consumption conditions raise the conditional left-tail quantiles and thereby reduce downside tail risk, whereas speculative shocks have only a modest effect.

On the right tail (q_{95} and q_{99}), macroeconomic shocks remain the most influential but now load negatively (-0.5028 and -1.2975). This pattern indicates that during economic expansions, the upside of returns is compressed, consistent with reduced uncertainty in favorable macro environments. In contrast, supply shocks as well as speculative and consumption shocks show no significant influence in the far right tail.

The quantile regressions highlight asymmetries in the transmission of structural shocks across different segments of the return distribution. Macroeconomic conditions mitigate downside risk but also dampen upside extremes. Supply shocks exert consistent downward pressure on central returns, yet their effects diminish to insignificance in the tails. Consumption shocks consistently lift lower-tail return quantiles, while speculative shocks contribute positively at the center of the distribution but have limited impact in the tails. These results show that the effects of structural shocks are heterogeneous across the return distribution, implying that tail-risk assessment should be conditioned on the nature of the underlying shocks.

5.5 Jump Variation in VaR

Tables ?? and ?? report the Value-at-Risk (VaR) evaluation at the 1% and 5% significance levels for both the left and right tails. We compare four models: Historical Simulation, GARCH, the standard asymmetric CAViaR, and an extended CAViaR specification augmented with lagged jump variation (denoted as CAViaR-JV). The tables report exceedance rates, the Kupiec unconditional coverage test, the Christoffersen conditional coverage test, and the Diebold–Mariano (DM) statistic. The DM statistic measures predictive loss *relative to the baseline CAViaR model*, with positive values indicating improved accuracy and negative values indicating worse performance.

[Insert Table ?? about here.]

[Insert Table ?? about here.]

For the left tail at the 1% level, both CAViaR and CAViaR-JV achieve nearly exact coverage rates (0.0100 and 0.0103). The Kupiec test does not reject correct calibration, while the Christoffersen test detects clustering of exceedances for both models (11.51*** and 6.72***). At the 5% left tail, both models again match the nominal rate (0.0501 and 0.0496). The Christoffersen test rejects for CAViaR (3.89**), but not for CAViaR-JV (1.42), indicating that including jump variation helps mitigate violation clustering in the left tail.

On the right tail, both CAViaR specifications show accurate unconditional coverage at the 5% and 1% levels, with Kupiec statistics insignificant in all cases. The Christoffersen test is insignificant at 5% (0.12 and 0.36) and marginally significant at 1% (3.38* and 3.22*), suggesting mild dependence of exceedances in the far upper tail.

Three out of four evaluation criteria indicate that incorporating jump components enhances VaR performance. On the left tail at the 1% level, CAViaR-JV underperforms the CAViaR benchmark ($DM = 10.54$), while Historical and GARCH achieve better relative accuracy ($DM = 0.35$ and 12.96 , respectively). At the 5% left tail, by contrast, CAViaR-JV delivers superior performance ($DM = 3.99$) and shows an additional advantage in conditional coverage. This pattern suggests that the most extreme 1% losses are primarily driven by new, event-specific discontinuities that lagged jump information cannot capture, whereas at the 5% quantile the persistence of jump activity and volatility clustering renders lagged jump variation sufficiently informative to enhance predictive accuracy.

For the right tail, the evidence is favorable for the jump-augmented model. At both the 5% and 1% levels, CAViaR-JV outperforms the baseline CAViaR, as reflected in strongly negative DM statistics (-9.85 and -21.54). By contrast, GARCH yields a positive and significant DM statistic at the 5% right tail (2.33), suggesting inferior predictive accuracy relative to the CAViaR benchmark, despite achieving acceptable unconditional coverage in some specifications.

These results highlight the asymmetric role of jump variation in risk forecasting, as lagged jump information improves accuracy for moderate tail risks through persistence and clustering effects. This finding is consistent with prior evidence (????) that incorporating jumps into volatility modeling and forecasting enhances risk assessment.

6 Conclusion

This paper examines how structural supply, demand, macroeconomic, and speculative shocks shape the dynamics of crude oil jumps and their implications for tail risk. Using high-frequency WTI futures data, we decompose realized variance into continuous and jump components and document that extreme oil price risks are fundamentally driven by jumps. These discontinuities cluster during crises and display marked directional asymmetries, underscoring the importance of distinguishing between positive and negative jump activity.

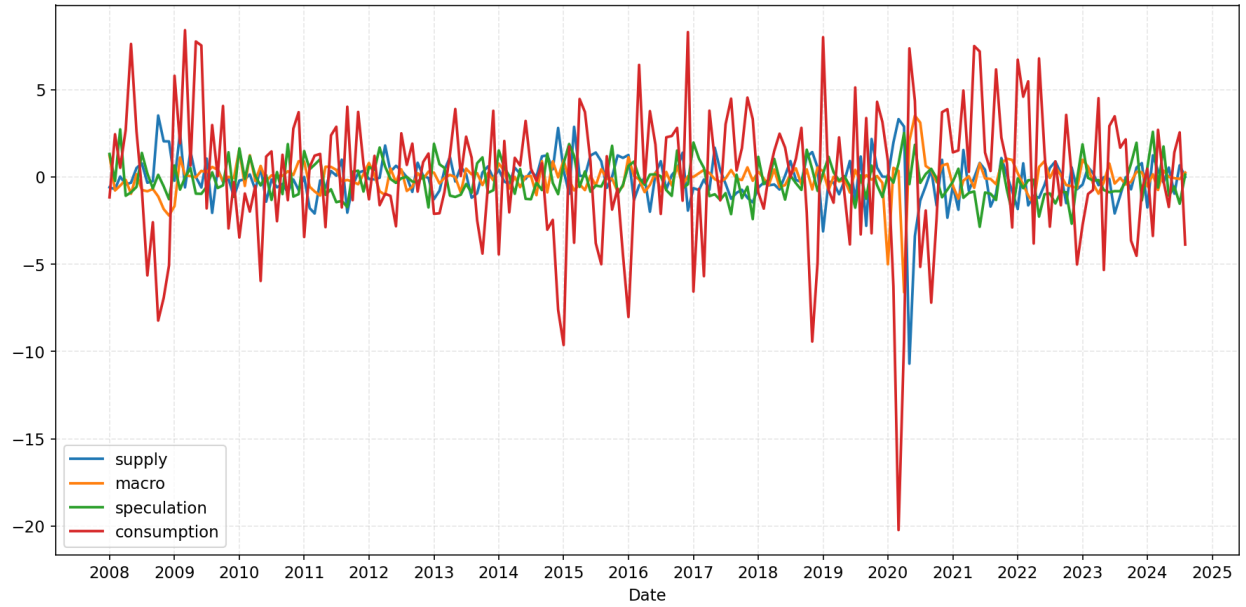
Our analysis shows that structural shocks leave distinct signatures on volatility components. Supply shocks amplify overall variance with particularly strong effects on positive jumps, while macroeconomic and consumption shocks reduce both continuous volatility and jump variation. Speculative shocks act asymmetrically, reducing average jump sizes more strongly on the downside while shifting intensity toward more frequent positive jumps. These results reveal that the origins of jumps are systematic rather than random and are closely tied to underlying economic fundamentals.

Quantile regressions highlight heterogeneity across the return distribution. Macroeconomic shocks mitigate downside extremes but compress upside returns, supply shocks depress the center yet vanish in the tails, consumption shocks stabilize the left tail, and speculative shocks primarily influence the center of the distribution. These findings emphasize that the transmission of shocks is highly state dependent and asymmetric across the conditional distribution of returns.

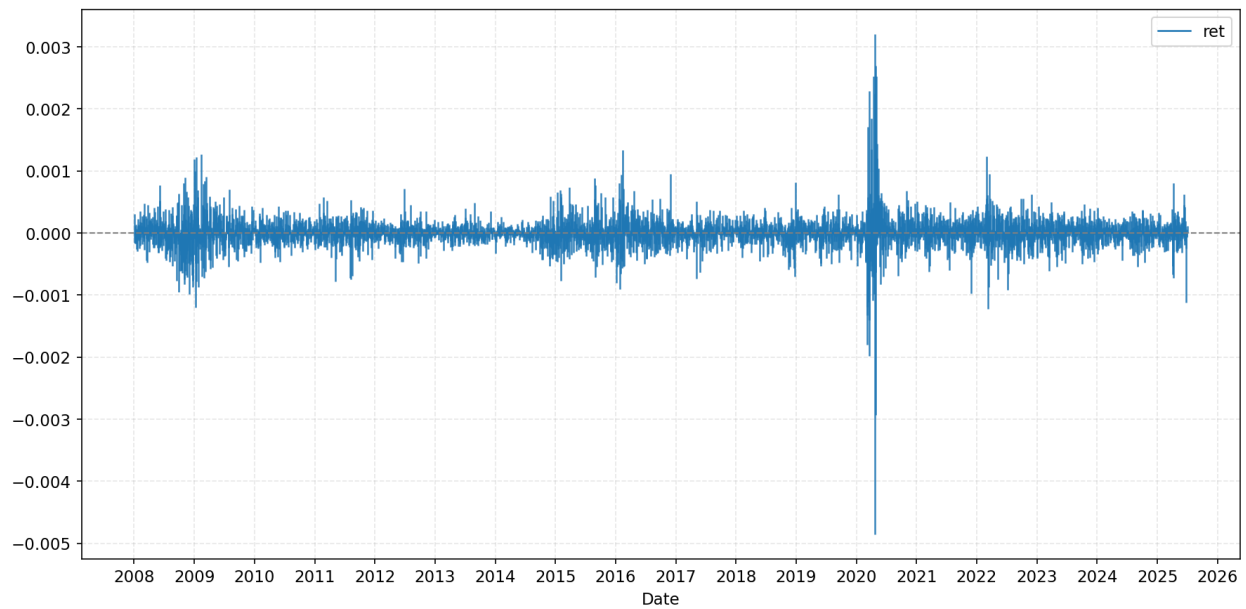
Finally, incorporating jump variation into quantile-based Value-at-Risk models improves predictive accuracy relative to standard approaches such as Historical Simulation, GARCH, and the baseline CAViaR model. Persistence and clustering effects render lagged jump information particularly useful in forecasting moderate tail risks and in improving conditional coverage.

This study highlights the role of structural shocks in shaping crude oil tail risk and demonstrate the value of embedding jump dynamics into both empirical analysis and risk management practice. By linking economic fundamentals to discontinuous price movements, this paper provides nuanced insights into the origins of oil market tail risks and their

implications for risk management.

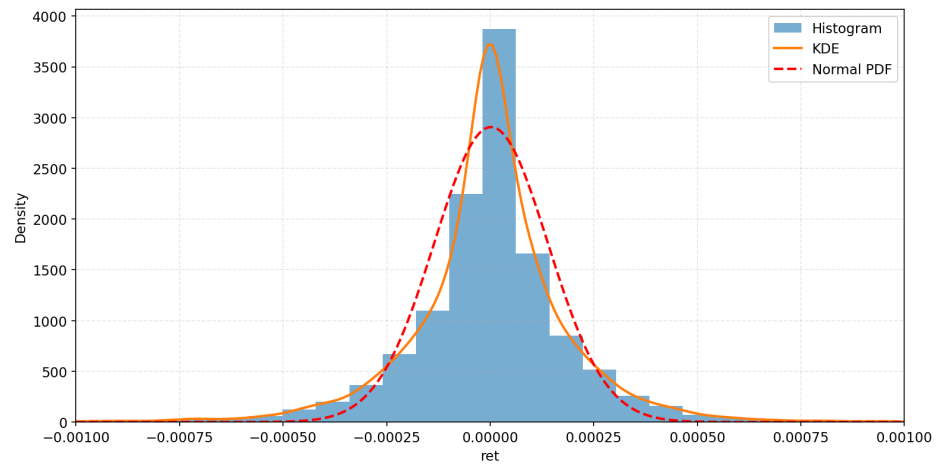


(a) Demand and Supply-driven Shocks

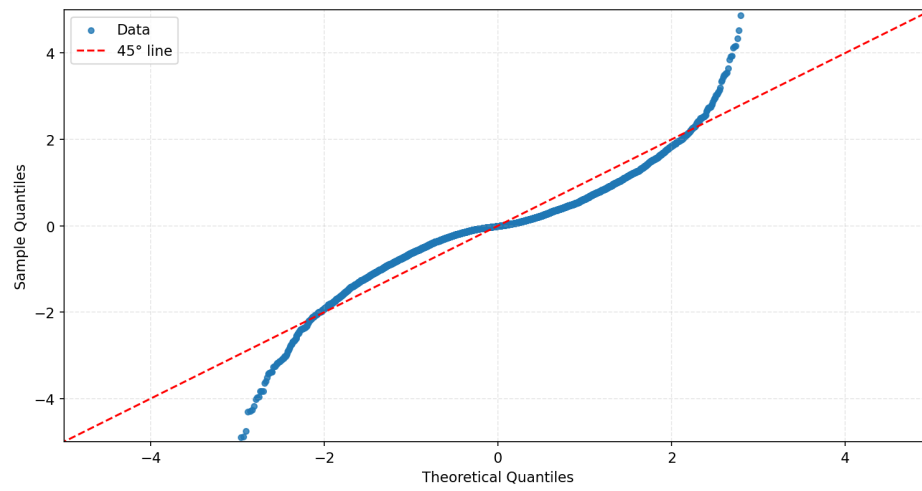


(b) Aggregated return

Figure 1: Demand and Supply-driven Shocks and Aggregated return

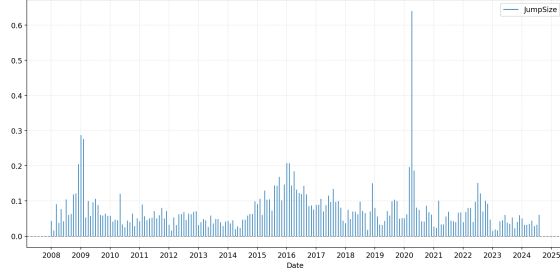


(a) Return Distribution

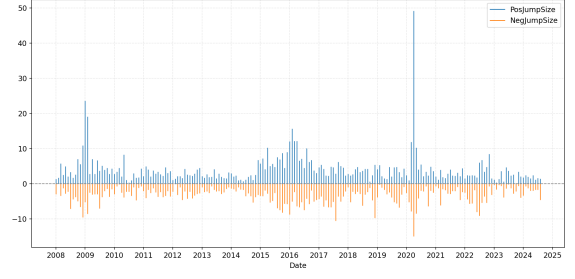


(b) QQ Plot

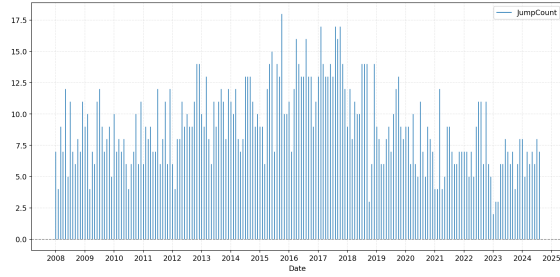
Figure 2: WTI Crude Oil Return Distribution



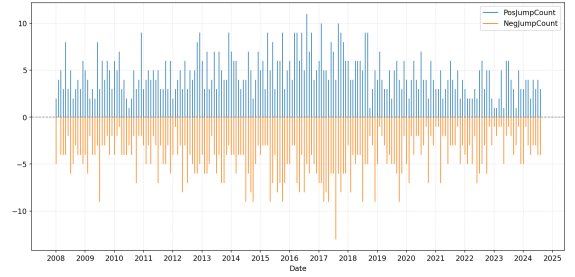
(a) Total jump size over time



(b) Positive and negative jump sizes



(c) Total jump count over time



(d) Positive and negative jump counts

Figure 3: Jump Size and Count Plots

Table 1: Summary Statistics for Returns in Major Shock Periods

	Mean	Volatility	Skewness	Kurtosis	Minimum	Maximum
Full Sample	0.0002	0.3316	0.6312	11.8337	-0.1217	0.1794
Global Financial Crisis	-0.0001	0.4655	0.1003	2.8069	-0.1082	0.1245
2014 Oil Price Collapse	-0.0015	0.3599	0.5578	3.1327	-0.0839	0.1085
2018 Trade War	0.0001	0.2455	-0.3073	2.6060	-0.0628	0.0689
COVID-19 Pandemic	0.0017	0.6682	0.3397	23.3848	-0.3539	0.2856
2022 Ukraine Crisis	-0.0000	0.3798	-0.0212	3.3460	-0.1105	0.1125

Note: This table reports the mean, volatility (standard deviation), skewness, kurtosis, minimum, and maximum of daily crude oil returns during the full sample and selected crisis periods.

Table 2: Summary Statistics for Oil Jump Events

Period	# Obs	# Jump Days	% Jump Days	# Pos Jump	% Pos Jump	# Neg Jump	% Neg Jump
Global Financial Crisis	547	141	25.7770	72	13.1627	69	12.6143
2014 Oil Price Collapse	639	232	36.3067	111	17.3709	121	18.9358
2018 Trade War	671	209	31.1475	106	15.7973	103	15.3502
COVID-19 Pandemic	686	158	23.0321	89	12.9738	69	10.0583
2022 Ukraine Crisis	393	88	22.3919	39	9.9237	49	12.4682

Note: Each row reports the number and percentage of jump days for major market periods. Positive and negative jump days and size distributions are reported separately.

Table 3: GPD Tail Parameter Estimates

Tail	q	Mean Exc.	Std. Exc.	Shape ξ	Scale σ
Returns					
Left	0.0250	0.0244	0.0451	0.3732	0.0151
Right	0.9750	0.0271	0.0457	0.5857	0.0128
Jump Size					
Left	0.0250	0.0064	0.0093	0.5762	0.0032
Right	0.9750	0.0238	0.0402	0.6800	0.0100

Note: This table reports Peaks-over-Threshold (GPD) estimates for return and jump tails on both the left and right sides. The estimated shape ξ and scale σ parameters are the heaviness and dispersion of the tails.

Table 4: Realized Variance Regression Coefficient Estimates

	$\Delta \log RV_t$	$\Delta \log BV_t$	$\Delta \log JV_t$	$\Delta \log RS_t^+$	$\Delta \log RS_t^-$	$\Delta \log JV_t^+$	$\Delta \log JV_t^-$
$Supply_t$	0.1008*** (3.35)	0.0995*** (3.26)	0.0952*** (2.79)	0.1051*** (3.66)	0.0950*** (3.00)	0.1205*** (3.58)	0.0944** (2.31)
$Macro_t$	-0.1167*** (-3.19)	-0.1154*** (-3.06)	-0.1636*** (-3.08)	-0.1145*** (-3.19)	-0.1219*** (-3.13)	-0.1621** (-2.37)	-0.1558*** (-3.08)
$Consumption_t$	-0.0605*** (-4.22)	-0.0599*** (-4.24)	-0.0653*** (-3.22)	-0.0604*** (-4.19)	-0.0613*** (-4.23)	-0.0656*** (-3.03)	-0.0679*** (-3.87)
$Speculation_t$	-0.0800*** (-2.59)	-0.0793*** (-2.62)	-0.0802* (-1.66)	-0.0912*** (-2.92)	-0.0700** (-2.23)	-0.1400*** (-2.86)	-0.0374 (-0.86)
$\Delta \log RV_{t-1}$	-0.0886 (-1.63)						
$\Delta \log BV_{t-1}$		-0.0970* (-1.73)					
$\Delta \log JV_{t-1}$			-0.2032*** (-3.59)				
$\Delta \log RS_{t-1}^+$				-0.0525 (-0.98)			
$\Delta \log RS_{t-1}^-$					-0.1429*** (-2.59)		
$\Delta \log JV_{t-1}^+$						-0.1586** (-2.40)	
$\Delta \log JV_{t-1}^-$							-0.3556*** (-6.56)

Note: This table reports coefficient estimates from ARX(1) regressions of first-differenced log realized measures on structural shocks. The dependent variables are $\Delta \log RV_t$, $\Delta \log BV_t$, $\Delta \log JV_t$, and the sign-decomposed semivariances $\Delta \log RS_t^+$, $\Delta \log RS_t^-$, $\Delta \log JV_t^+$, and $\Delta \log JV_t^-$. HAC-robust t -statistics based on Newey–West standard errors are reported in parentheses. Statistical significance at the 10%, 5%, and 1% levels is indicated by *, **, and ***, respectively.

Table 5: Jump Regression Coefficient Estimates

	Size		Count	
	PosJump	NegJump	PosJump	NegJump
<i>Supply_t</i>	-0.1018 (-0.95)	-0.0951 (-0.89)	-0.3975*** (-4.60)	0.2462*** (3.18)
<i>Macro_t</i>	-0.2378 (-1.56)	-0.1110 (-0.67)	-0.5446*** (-4.31)	-0.1522 (-1.59)
<i>Consumption_t</i>	-0.0348 (-0.87)	-0.0768** (-2.44)	0.0293 (1.02)	0.0783*** (2.64)
<i>Speculation_t</i>	-0.2197** (-2.15)	-0.2946*** (-3.13)	0.7079*** (7.03)	-1.3571*** (-13.03)

Note: This table reports coefficient estimates from jump regression models. Size regressions are estimated using OLS on the logarithm of absolute positive and negative jump sizes, with Newey–West HAC-robust t -statistics reported in parentheses. Count regressions are estimated using Negative Binomial models for positive and negative jump counts, with GLM-based test statistics reported in parentheses. Statistical significance at the 10%, 5%, and 1% levels is denoted by *, **, and ***, respectively.

Table 6: Quantile Regression Coefficient Estimates

	q_{01}	q_{05}	q_{10}	q_{50}	q_{90}	q_{95}	q_{99}
<i>Intercept</i>	-6.2305*** (-36.07)	-3.6986*** (-54.19)	-2.5899*** (-54.31)	0.0571** (2.50)	2.5361*** (62.11)	3.5295*** (49.78)	6.1556*** (31.17)
r_{t-1}	0.0315 (0.19)	-0.0344 (-0.86)	-0.0052 (-0.21)	-0.0722*** (-8.00)	-0.0810*** (-3.41)	-0.0643 (-1.27)	-0.0571 (-0.30)
<i>Supply</i> $_{t-1}$	0.2118 (0.77)	0.0240 (0.28)	-0.0699 (-1.35)	-0.1121*** (-5.45)	-0.1096** (-2.28)	-0.1661 (-1.63)	-0.3367 (-0.80)
<i>Macro</i> $_{t-1}$	1.0087** (2.20)	0.5155*** (3.60)	0.2513*** (2.95)	0.0744** (2.31)	-0.2251*** (-2.96)	-0.5028*** (-3.21)	-1.2975** (-1.96)
<i>Consumption</i> $_{t-1}$	0.2526*** (2.73)	0.1710*** (5.94)	0.1441*** (7.89)	0.0773*** (10.53)	0.0246 (1.59)	0.0074 (0.26)	-0.0781 (-0.87)
<i>Speculation</i> $_{t-1}$	0.3030 (1.46)	0.1335* (1.74)	0.1510*** (2.90)	0.0954*** (4.10)	-0.0094 (-0.22)	0.0012 (0.02)	0.1990 (0.90)

Note: Conditional quantiles of daily returns are estimated at $q = \{0.01, 0.05, 0.10, 0.50, 0.90, 0.95, 0.99\}$. Regressors include an intercept, the lagged return r_{t-1} , and lagged structural shocks (one-day lag; monthly shocks forward-filled to the daily index). Reported are coefficient estimates with robust t -statistics in parentheses. Statistical significance at the 10%, 5%, and 1% levels is denoted by *, **, and ***, respectively.

Table 7: VaR Performance on left tail

	Exceedance Rate	Kupiec	Christoffersen	DM vs CAViaR
<i>VaR</i> _{1%}				
Historical	0.0174	20.9413*** (0.0000)	3.9562** (0.0467)	-0.3467 (0.7288)
Garch	0.0217	43.5379*** (0.0000)	0.5241 (0.4691)	-12.9553*** (0.0000)
CAViaR	0.0100	0.0002 (0.9876)	11.5094*** (0.0007)	
CAViaR-JV	0.0103	0.0289 (0.8650)	6.7176*** (0.0095)	10.5387*** (0.0000)
<i>VaR</i> _{5%}				
Historical	0.0482	0.0125 (0.9111)	3.8834** (0.0488)	-2.7395*** (0.0062)
Garch	0.0551	2.2510 (0.1335)	0.5415 (0.4618)	-1.8985* (0.0577)
CAViaR	0.0501	0.0013 (0.9717)	3.8871** (0.0487)	
CAViaR-JV	0.0496	0.0113 (0.9152)	1.4162 (0.2340)	-3.9896*** (0.0001)

Note: Value-at-Risk performance is evaluated at $\alpha \in \{1\%, 5\%\}$ for the left tails. Reported metrics are the empirical exceedance rate, the Kupiec test (unconditional coverage), the Christoffersen test (conditional coverage), and the Diebold–Mariano (DM) statistic relative to the baseline CAViaR model. Each statistic is shown on the first line with its p -value in parentheses on the second line. For DM we use the loss differential $DM_t = \ell_t^M - \ell_t^B$ where ℓ_t^M and ℓ_t^B are the model and benchmark losses. $DM > 0$ indicates higher predictive loss than the benchmark and $DM < 0$ indicates lower predictive loss. Statistical significance at the 10%, 5%, and 1% levels is denoted by *, **, and ***, respectively.

Table 8: VaR Performance on right tail

	Exceedance Rate	Kupiec	Christoffersen	DM vs CAViaR
<i>VaR_{95%}</i>				
Historical	0.0504	0.2838	2.7487*	1.3560
		(0.5942)	(0.0973)	(0.1752)
Garch	0.0465	1.0803	0.5174	2.3274**
		(0.2986)	(0.4720)	(0.0200)
CAViaR	0.0499	0.0013	0.1220	
		(0.9717)	(0.7269)	
CAViaR-JV	0.0501	0.0013	0.3600	-9.8518***
		(0.9717)	(0.5485)	(0.0000)
<i>VaR_{99%}</i>				
Historical	0.0189	28.5348***	5.4513**	-1.3821
		(0.0000)	(0.0196)	(0.1670)
Garch	0.0146	7.7098***	0.0432	-17.5567***
		(0.0055)	(0.8353)	(0.0000)
CAViaR	0.0098	0.0197	3.3753*	
		(0.8885)	(0.0662)	
CAViaR-JV	0.0100	0.0002	3.2165*	-21.5374***
		(0.9876)	(0.0729)	(0.0000)

Note: Value-at-Risk performance is evaluated at $\alpha \in \{1\%, 5\%\}$ for the right tails. Reported metrics are the empirical exceedance rate, the Kupiec test (unconditional coverage), the Christoffersen test (conditional coverage), and the Diebold–Mariano (DM) statistic relative to the baseline CAViaR model. Each statistic is shown on the first line with its p -value in parentheses on the second line. For DM we use the loss differential $DM_t = \ell_t^M - \ell_t^B$ where ℓ_t^M and ℓ_t^B are the model and benchmark losses. $DM > 0$ indicates higher predictive loss than the benchmark and $DM < 0$ indicates lower predictive loss. Statistical significance at the 10%, 5%, and 1% levels is denoted by *, **, and ***, respectively.